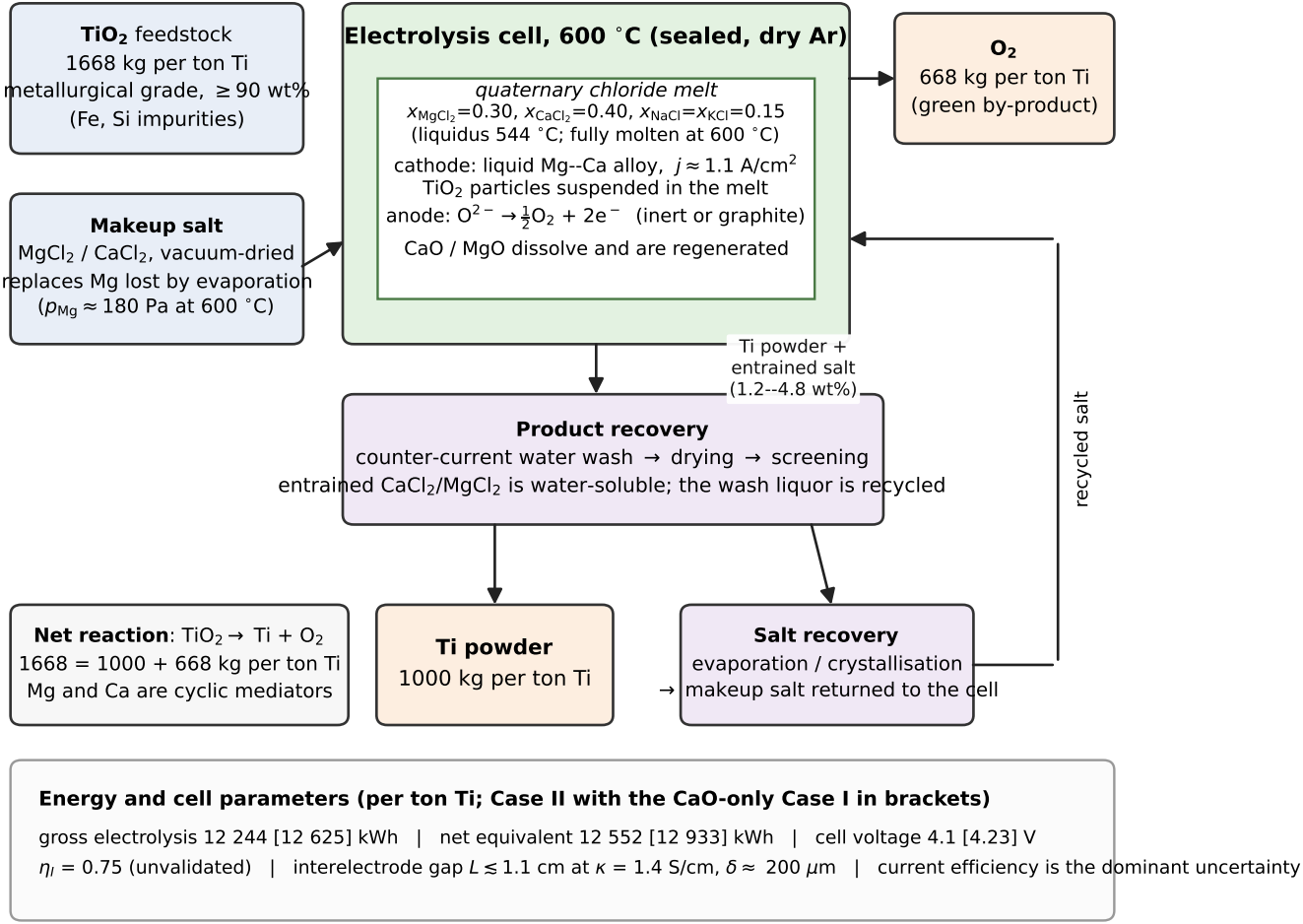

Energy and cell parameters (per ton Ti; Case II with the CaO-only Case I in brackets)

gross electrolysis 12 244 [12 625] kWh | net equivalent 12 552 [12 933] kWh | cell voltage 4.1 [4.23] V

η_I = 0.75 (unvalidated) | interelectrode gap $L \lesssim 1.1$ cm at κ = 1.4 S/cm, $\delta \approx$ 200 μm | current efficiency is the dominant uncertainty